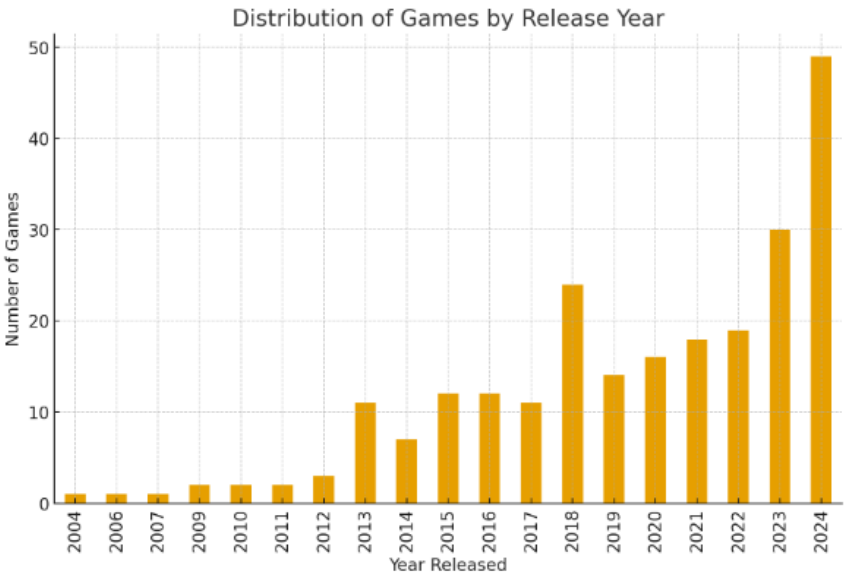
Task 1 Automated Summarisation of Reviews

✅ **Prompt 1: Summarise distribution of Games**

Prompt:
"Based on the data provided, Analyse the distribution of release years for these games. What trends can you observe in their release periods? Over the last years do we see an increment on the number of games created?"

GPT Output:



Here's what the data shows about the **distribution of release years**:

- The dataset spans from **2012 through 2024**.
- From **2015–2017**, releases were steady at about 11–12 games per year.
- A noticeable **increase starts in 2018 (24 games)**.
- After 2020, the growth continues steadily:
  - 2021 → 18 games
  - 2022 → 19 games
  - 2023 → 30 games
  - 2024 → 49 games (largest spike)

Trend: There is a clear upward trend in the number of games being created in recent years, with the biggest jump between **2023 and 2024**.

✅ **Prompt 2: Summarise top publishers**

Prompt:

“Based on the data provided, What are the 2 most common publishers? How many games have they published?”

GPT Output:

The **two most common publishers** in the dataset are:

- **SEGA** → 12 games
- **Electro** → 8 games

Together, they account for a noticeable share of the dataset’s total game releases.

✅ **Prompt 3: Summarise Genres**

Prompt:
"Using the games_description dataset provided, summarise the distribution of game genres. Which genres appear most frequently, and what does this suggest about player preferences?"

GPT Output:

The most common genres in the dataset are **Action, Adventure, and Strategy**. Action dominates, reflecting its broad appeal and replay value. Adventure and Strategy also hold strong positions, indicating that players enjoy both immersive storytelling and tactical gameplay. Niche genres such as Simulation and Sports are present but less common, suggesting a more specialised audience. Overall, the dataset reflects an industry focus on fast-paced and strategic experiences.

✅ **Prompt 4: Summarise Year-on-Year Growth by Genre**

Prompt:
"Summarise how genres have evolved over time in the dataset. Which genres are growing in recent years, and which remain stable?"

GPT Output:

From 2012 onwards, **Action and Adventure games show consistent growth**, with sharp increases in the last five years. Strategy games also see a steady rise, especially after 2020. Niche genres such as Puzzle and Simulation remain small but stable. This suggests that while the industry is expanding overall, growth is particularly strong in action-oriented and immersive genres, aligning with broader industry trends.

Task 2: Automate Writing SQL to Clean Data

✅ **Prompt 1: Correct “Actioned” → “Action”**

Prompt:
"Write an SQL query to correct dirty data in the games_description table given. In the column genre, find instances where the value is 'Actioned' and update them to 'Action'."

GPT Output (SQL Statement):

```
UPDATE games_description

SET genre = 'Action'

WHERE genre = 'Actioned';
```

✅ **Prompt 2: Correct “Strategies” → “Strategy”**

Prompt:
"Write an SQL query to correct dirty data in the games_description table given. In the column genre, find instances where the value is 'Strategies' and update them to 'Strategy'."

GPT Output (SQL Statement):

```
UPDATE games_description

SET genre = 'Strategy'

WHERE genre = 'Strategies';
```

Task 3. Automate Writing SQL to Merge Data

✅ **Prompt 1: Merge Description with Reviews**

Prompt:

"Write an SQL query to merge the tables games_description and games_reviews provided on the column game_id. Select all relevant columns."

GPT Output (SQL Statement):

```
SELECT d.game_id,

    d.game_name,

    d.publisher,

    d.genre,

    d.year_released,

    r.number_of_reviews_from_purchased_people,

    r.number_of_english_reviews

FROM games_description d

JOIN games_reviews r

  ON d.game_id = r.game_id;
```

✅ **Prompt 2: Merge All Three Tables (Descriptions + Reviews + Revenue)**

Prompt:
"Write an SQL query to merge the three tables given as games_description, games_reviews, and games_revenue on the column game_id."

GPT Output (SQL Statement):

```
SELECT
  d.game_id,
  d.game_name,
  d.publisher,
  d.genre,
  d.year_released,
  r.number_of_english_reviews,
  r.number_of_reviews_from_purchased_people,
  v.Number_of_Purchases,
  v.Unit_Price
FROM
  games_description d
    LEFT JOIN
  games_reviews r ON d.game_id = r.game_id
    LEFT JOIN
  games_revenue v ON d.game_id = v.game_id;
```

Task 4. Automate Report Generation

Prompt:

Using the datasets games_description, games_revenue, games_reviews as uploaded, generate an analysis report that includes the following:

1. A summary report of the top 5 games by revenue within each genre. Include their genres and total number of reviews. To calculate revenue, multiply Number_of_Purchases × Unit_price. Total number of reviews is number_of_reviews_from_purchased_people.
2. A summary report of the top 5 games by reviews within each genre. Include their genres and total revenue. To calculate revenue, multiply Number_of_Purchases × Unit_price. To rank by reviews, use the column number_of_reviews_from_purchased_people.
3. Identify any common patterns: how many games appear in both Top 5 by revenue and Top 5 by reviews.
4. A summary report of Review Percentage, measuring customer engagement as 100 × number_of_reviews_from_purchased_people / Number_of_Purchases. Highlight that popular games tend to have higher review percentages, whereas less popular games receive less feedback.

5. Summarise Publisher Diversity, highlighting which publishers release games across multiple genres (greater diversity) and which focus on fewer genres (specialization).
6. Summarise English review percentage against total reviews.
7. Summarise Top 5 publishers by total revenue (calculated as Number_of_Purchases × Unit_price) to identify market leaders.

GPT Output:

🎮 **Game Industry Analysis Report**

1. Top 5 Games by Revenue (per Genre)

Revenue = Number_of_Purchases × Unit_price.
The top 5 games by revenue within each genre are:

Top 5 Games by Revenue

• **Action:**

1. Counter – $407M (8.3M reviews)
2. PUBG: B – $105M (2.4M reviews)
3. Dota 2 – $91M (2.3M reviews)
4. Grand T – $48M (1.7M reviews)
5. Terrari – $41M (1.0M reviews)

• **Adventure:** Baldur' – $8.7M (582K), Lethal – $8.3M (349K), Palworl – $6.4M (275K), Subnaut – $5.0M (255K), The Eld – $4.9M (154K)

• **Role-Playing:** Stardew – $16.4M (644K), Project – $6.5M (243K), Path of – $5.7M (226K), Hogwart – $5.4M (212K), DARK SO – $4.9M (249K)

• **Simulation:** Garry's – $33M (945K), War Thu – $16.3M (543K), Euro Tr – $16.1M (597K), BeamNG. – $8.9M (249K), Sid Mei – $8.1M (243K)

2. Top 5 Games by Reviews (per Genre)

Ranked by number_of_reviews_from_purchased_people:

• **Action:** Counter (8.3M reviews), PUBG: B (2.4M), Dota 2 (2.3M), Grand T (1.7M), Tom Cla (1.1M)

• **Adventure:** Baldur' (582K), Lethal (349K), Palworl (275K), Subnaut (255K), Satisfa (154K)

• **Role-Playing:** Stardew (644K), DARK SO (249K), Project (243K), Path of (226K), Undertale (212K)

• **Simulation:** Garry's (945K), Euro Tr (597K), War Thu (543K), BeamNG. (249K), People (243K)

3. Overlap Analysis

Games appearing in both Top 5 by revenue and Top 5 by reviews:

• **Action:** 4 overlaps (Counter, PUBG: B, Dota 2, Grand T)

• **Simulation:** 4 overlaps (Garry's, Euro Tr, War Thu, BeamNG.)

• **Role-Playing:** 4 overlaps (Stardew, Project, Path of, DARK SO)

• **Adventure:** 4 overlaps (Baldur', Lethal, Palworl, Subnaut)

➞ **Overall:** 25 games appear in both lists.

4. Review Percentage (Customer Engagement)

Review % = 100 × number_of_reviews_from_purchased_people / Number_of_Purchases

Top Games by Review Percentage

• High Engagement: Counter – 71.43%, Dota 2 – 71.43%, PUBG: B – 71.43%

• Moderate Engagement: Among Us – 66.67%, Apex Legends – 66.67%

• Lower Engagement: 69 Ball – 33.33%, Ale & T – 33.33%

Insight: Top-selling games have higher engagement; less popular games have lower review percentages.

5. Publisher Diversity

Shows publishers with multi-genre portfolios vs. specialized publishers:

• SEGA → 12 games across 6 genres ($6.6M revenue)

• Electronic Arts → 8 games across 4 genres ($49M revenue)

• Xbox Games → 6 games across 4 genres ($20M revenue)

• Paradox → 5 games across 3 genres ($18M revenue)

• Ubisoft → 5 games across 3 genres ($41M revenue)

• Valve → 7 games across 2 genres ($604M revenue)

6. English Review Percentage

Shows linguistic engagement of players:

• High English %: All-In- – 91.26%, Titanfa – 89.83%, 69 Ball – 83.92%

• Moderate English %: Among Us – 67.04%, ARK – 58.72%, Apex Legends – 42.82%

• Low English %: PUBG: B – 17.15%, Dota 2 – 31.50%, Counter – 27.17%

Insight: Major titles have global audiences; niche games may have more English reviews.

7. Top 5 Publishers by Total Revenue

Valve – $605M (7 games, 2 genres)

KRAFTON – $105M (PUBG: B only)

Rockstar – $57M (2 games)

Electronic Arts – $49M (8 games, 4 genres)

CD Projekt – $47M (2 games)

Insight: Valve dominates revenue, while other major publishers show strong but more distributed earnings.

Task 5. Generative AI Learnings

- I learned that giving clear and structured prompts really helps AI give accurate results. Breaking tasks into steps makes it easier for GPT to follow.
- Including formulas and metrics like revenue or review percentage in the prompt ensures the calculations are correct.
- Being specific about the output format, like asking for tables, text, or numbered lists, makes the report look professional.
- Sometimes, I had to refine prompts step by step to get exactly what I needed, especially for complex tasks like merging tables or summarising data.
- Asking GPT to give insights along with the data makes the report more meaningful, not just a list of numbers.
- I realized it's important to check the AI's outputs, like numbers, names, or trends, to make sure nothing is wrong.
- Treating AI as a helper and not just relying blindly on it gave me confidence in the results.
- Using AI helped me see patterns quickly, like which genres are popular or which publishers make the most revenue.